

BG770A-GL&BG95xA-GL

PSM Application Note

LPWA Module Series

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About the Document

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-	2021-01-29	Igor STANCIC	Creation of the document
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1 Introduction

This document describes the Power Saving Mode (PSM) feature and explains how to use the PSM function on the Quectel BG770A-GL, BG950A-GL and BG951A-GL modules.

The PSM feature enables IoT devices to stay inactive or powered down most of the time to save power. The PSM function wakes up the device only during data transmission, which usually happens for a short period of time. The PSM feature is especially important for eMTC/NB-IoT devices and applications that have the following characteristics:

- The eMTC/NB-IoT devices and applications are frequently inactive
- Their active communication is of short duration
- Data mainly originate from eMTC/NB-IoT devices (although the eMTC/NB-IoT devices can also download data when they are active)
- There are power constraints (the devices are running on a battery)
- The eMTC/NB-IoT devices require long battery life

The PSM mode defined by 3GPP Release 13 is similar to power-off, but the UE (i.e., the BG770A-GL, BG950A-GL and BG951A-GL modules) remains registered on the network, so there is no need to re-attach or re-establish PDN connections. Therefore, the UE in PSM is not immediately available to mobile terminating services. The UE using PSM is available to mobile terminating services when it is in connected mode and during an Active Time that comes after the connected mode. The connected mode is caused by a mobile originated event like data transmission or signalling, e.g. after a periodic TAU procedure. The PSM is therefore suitable for UE that is expecting only infrequent mobile originating and terminating services, which can accept a corresponding latency in the mobile terminating communication.

NOTE

For a more detailed description of the PSM mode defined by 3GPP Release 13, see **document [1]**.

2 Description of PSM AT Commands

2.1. AT Command Introduction

2.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

2.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding Write Command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

2.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

2.3. Description of AT Commands

2.3.1. AT+CPSMS Power Saving Mode Setting

The Write Command configures the PSM parameters of a module. It controls:

- whether the module wants to apply PSM or not,
- the requested extended periodic TAU value in E-UTRAN, and
- the requested Active Time value.

See the URCs provided by **AT+CEREG** for the Active Time value and the extended periodic TAU value that are allocated to the module by the network in E-UTRAN.

A special form of the command can be given as **AT+CPSMS=** (with all parameters omitted). In this form, the use of PSM is disabled (**<mode>=0**) and data for all parameters in the command are removed.

The Read Command returns the current parameter values.

The Test Command returns the supported **<mode>s**, the requested extended periodic TAU value in E-UTRAN and the requested Active Time value as compound values.

AT+CPSMS Power Saving Mode Setting	
Test Command AT+CPSMS=?	Response +CPSMS: (list of supported <mode>s),,,(list of supported <Requested_Periodic-TAU>s),(list of supported <Requested_Active-Time>s) OK
Read Command AT+CPSMS?	Response +CPSMS: <mode>,,,[<Requested_Periodic-TAU>],[<Requested_Active-Time>] OK

Write Command AT+CPSMS=[<mode>,,[,<Requested_Periodic-TAU>[,<Requested_Active-Time>]]]	Response OK If there is any other error: ERROR
Maximum Response Time	4000 ms
Characteristics	The command takes effect immediately. The configurations will be saved automatically.
Reference 3GPP TS 27.007	

Parameter

<mode>	Integer type. Disable or enable the use of PSM in the UE. 0 Disable the use of PSM 1 Enable the use of PSM
<Requested_Periodic-TAU>	String type. One byte in an 8-bit format. Requested extended periodic TAU value (T3412) to be allocated to the UE in E-UTRAN (e.g. "00001010" equals 100 minutes). In the Write Command, when <mode> is 1 and <requested_periodicTAU> is omitted, the value that was configured last is used. Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes 0 0 1 value is incremented in multiples of 1 hour 0 1 0 value is incremented in multiples of 10 hours 0 1 1 value is incremented in multiples of 2 seconds 1 0 0 value is incremented in multiples of 30 seconds 1 0 1 value is incremented in multiples of 1 minute
<Requested_Active-Time>	String type. One byte in an 8-bit format. Requested Active Time value (T3324) to be allocated to UE (e.g., "00001111" equals 1 minute). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 2 seconds 0 0 1 value is incremented in multiples of 1 minute 0 1 0 value is incremented in multiples of decihours 1 1 1 value indicates that the timer is deactivated.

Example

```
AT+CPSMS=1,,"0000100","0001111" //Set the requested T3412 value to 40 minutes, and set the
OK                               requested T3324 value to 30 seconds.
```

WARNING

After entering PSM, all digital logic is powered down, therefore, it is prohibited to provide any external voltage to the module's I/O ports that are not defined as a wake-up source. VDD_EXT pin can be used to monitor PSM activity and control the external logic.

NOTE

If the debugging mode is enabled (**AT+QCFGEXT="debug"**), depending on the logging activity load, the module can refuse to enter PSM mode.

2.3.2. AT+QPSMS Extended Power Saving Mode Setting

This is an extended AT command, developed by Quectel, for setting PSM parameters. A special form of the command can be given as **AT+QPSMS=** (with all parameters omitted). In this form, the use of PSM is disabled (**<mode>=0**) and data for all parameters in the command are removed. It is similar to **AT+CPSMS**.

AT+QPSMS Extended Power Saving Mode Setting	
Test Command AT+QPSMS=?	Response +QPSMS: (list of supported <mode>s),,(list of supported <Requested_Periodic-TAU>s),(list of supported <Requested_Active-Time>s) OK
Read Command AT+QPSMS?	Response +QPSMS: <mode> ,,,[<Network_Periodic-TAU>],[<Network_Active-Time>] OK
Write Command AT+QPSMS=[<mode>,,[<Requested_Periodic-TAU>],[<Requested_Active-Time>]]]	Response OK If there is any other error: ERROR
Maximum Response Time	4000 ms
Characteristics	The command takes effect immediately. The configurations will be saved automatically.

Reference
3GPP TS 27.007

Parameter

<mode>	Integer type. Disable or enable the use of PSM in UE. 0 Disable the use of PSM 1 Enable the use of PSM
<Requested_Periodic-TAU>	String type. One byte in an 8-bit format. Requested extended periodic TAU value (T3412) to be allocated to UE in E-UTRAN (e.g., "00001010" equals 100 minutes). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes 0 0 1 value is incremented in multiples of 1 hour 0 1 0 value is incremented in multiples of 10 hours 0 1 1 value is incremented in multiples of 2 seconds 1 0 0 value is incremented in multiples of 30 seconds 1 0 1 value is incremented in multiples of 1 minute
<Requested_Active-Time>	String type. One byte in an 8-bit format. Requested Active Time value (T3324) to be allocated to UE (e.g., "00001111" equals 1 minute). Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 define the timer value unit as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 2 seconds 0 0 1 value is incremented in multiples of 1 minute 0 1 0 value is incremented in multiples of decihours 1 1 1 value indicates that the timer is deactivated.
<Network_Periodic-TAU>	Integer type. Extended periodic TAU value (T3412) to be allocated to UE in E-UTRAN. The value is specified by the network.
<Network_Active-Time>	Integer type. Active timer value (T3324) to be allocated to UE in E-UTRAN. The value is specified by the network.

Example

```
AT+QPSMS=1,,"00000100","00001111" //Set the requested T3412 value to 40 minutes, and set the
OK                                     requested T3324 value to 30 seconds.
AT+QPSMS?                             //Query the PSM mode, the periodic TAU and the active
                                         time specified by the network.
```

```
+QPSMS:1,,,"86400","2"
OK
```

WARNING

After entering PSM, all digital logic is powered down, therefore, it is prohibited to provide any external voltage to the module's I/O ports that are not defined as a wake-up source. VDD_EXT pin can be used to monitor PSM activity and control the external logic.

NOTE

If the debugging mode is enabled (**AT+QCFGEXT="debug"**), depending on the logging activity, the module can refuse to enter PSM mode.

2.3.3. AT+QPSMCFG PSM Feature and Minimum Threshold Value Setting

The Write Command enables or disables the PSM feature and sets the minimum threshold value of the PSM cycle. A special form of the command can be given as **AT+QPSMCFG=** (with all parameters omitted). In this form, both parameters are set to their specific default values.

AT+QPSMCFG PSM Feature and Minimum Threshold Value Setting

Test Command AT+QPSMCFG=?	Response +QPSMCFG: (range of supported <threshold>s),(range of supported <PSM_version>s) OK
Write Command AT+QPSMCFG=[<threshold>[,<PSM_version>]]	Response OK If there is any error: ERROR
Read Command AT+QPSMCFG?	Response +QPSMCFG: <threshold>,<PSM_version> OK
Maximum Response Time	300 ms
Characteristics	The command takes effect after rebooting. The configurations will be saved automatically.

Parameter

<threshold>	Integer type. Minimum threshold value of the PSM cycle.
-------------	---

	Condition to enter PSM: <threshold> less than PSM cycle (= T3412 - T3324). Range: <u>20</u> -4294967295. Unit: second.
<PSM_version>	Integer type. Bitmask to indicate the PSM feature (1: Enable; 0: Disable). Each bit is configured independently. Range: 0- <u>2</u> -15. Bit 0 PSM without network coordination Bit 1 Rel-12 PSM without context retention Bit 2 Rel-12 PSM with context retention Bit 3 PSM in-between eDRX cycles

Example

```

AT+QPSMCFG=100           //Set the threshold to 100 seconds.
OK
AT+QPSMCFG?              //Query the threshold value and the PSM version.
+QPSMCFG: 100,4
OK

```

2.4. URC Description

2.4.1. +QPSMTIMER Indicate the TAU Duration and Active Time Duration

The URC is used to indicate the TAU duration and active time duration of a module's PSM. The URC is disabled by default and can be enabled with **AT+QCFG="psm/urc",1**.

+QPSMTIMER Indicate the TAU Duration and Active Time Duration

+QPSMTIMER: <TAU_timer>,<T3324_timer>	Indicate the TAU duration and active time duration of a module's PSM.
--	---

Parameter

<TAU_timer>	Integer type. The TAU duration of PSM. Unit: second.
<T3324_timer>	Integer type. The active time duration of PSM. Unit: second.

Example

```

AT+QCFG="psm/urc"
+QCFG: "psm/urc",0
OK
AT+QCFG="psm/urc",1
OK

```

+QPSMTIMER: 86400,2

//The TAU and active timer value of PSM.

3 Waking Up from PSM

There are three scenarios to wake up the module:

- Forced: Hold the PON_TRIG high to wake up the module.
- Drive PWRKEY low to wake up the module. The UART port stays inaccessible until the PON_TRIG is pulled high.
- Automatic: When the T3412 timer expires, the module wakes up automatically. The UART port stays inaccessible until the PON_TRIG is pulled high.

3.1. Waking Up from PSM with the PON_TRIG Pin

To wake the module up from the PSM to realize communication between the module and the network.

- Step 1:** Hold the PON_TRIG high and check if the module has been woken up from PSM.
Step 2: Establish the active UL and/or DL communication.

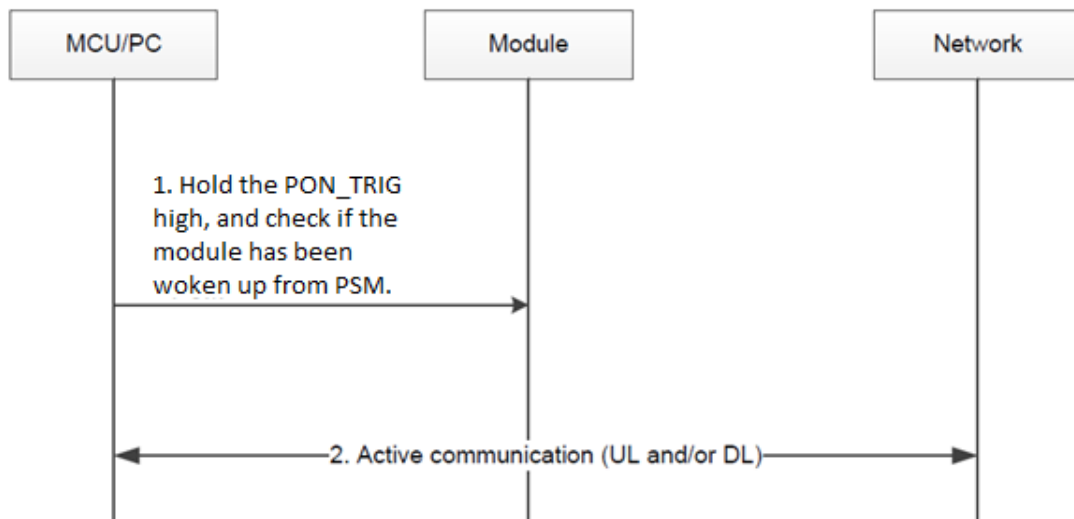


Figure 1: Waking Up the Module from PSM with PON_TRIG

3.2. Waking Up from PSM with the PWRKEY

Like the PON_TRIG solution, the module can be woken up from PSM through steps below to realize communication between the module and the network.

Step 1: Drive PWRKEY low and check if the module has been woken up from PSM. The UART port stays inaccessible until the PON_TRIG is pulled high.

Step 2: Establish the active UL and/or DL communication.

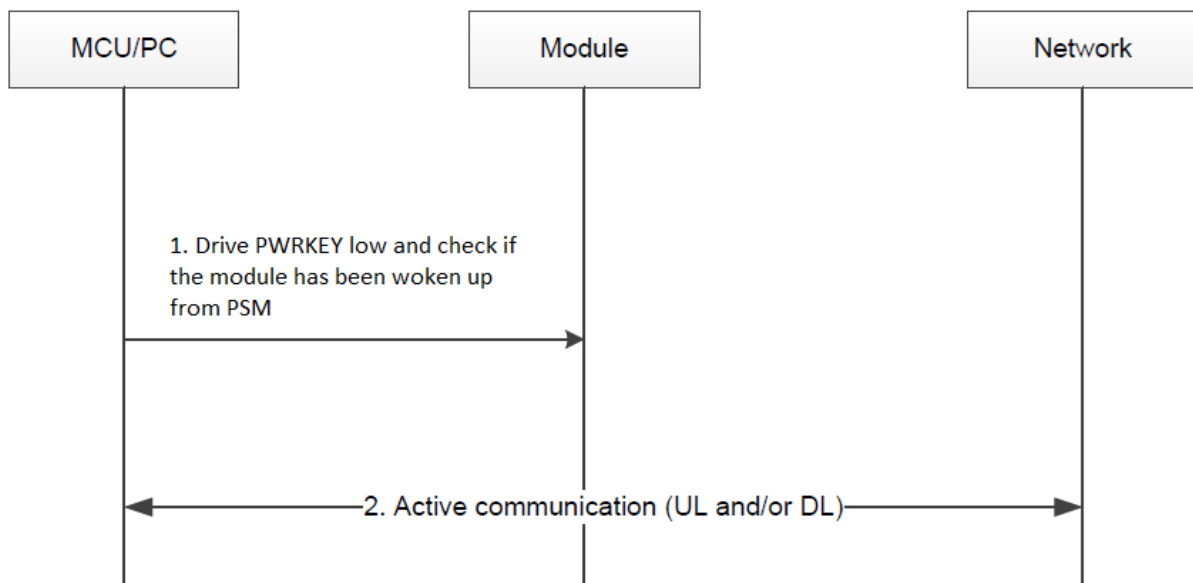


Figure 2: Waking up the Module from PSM with PWRKEY

3.3. Waking Up from PSM when the T3412 Timer Expires

When the T3412 (extended TAU timer) expires, the module automatically wakes up from PSM.

The following figure illustrates the automatic procedure for waking up a module from PSM.

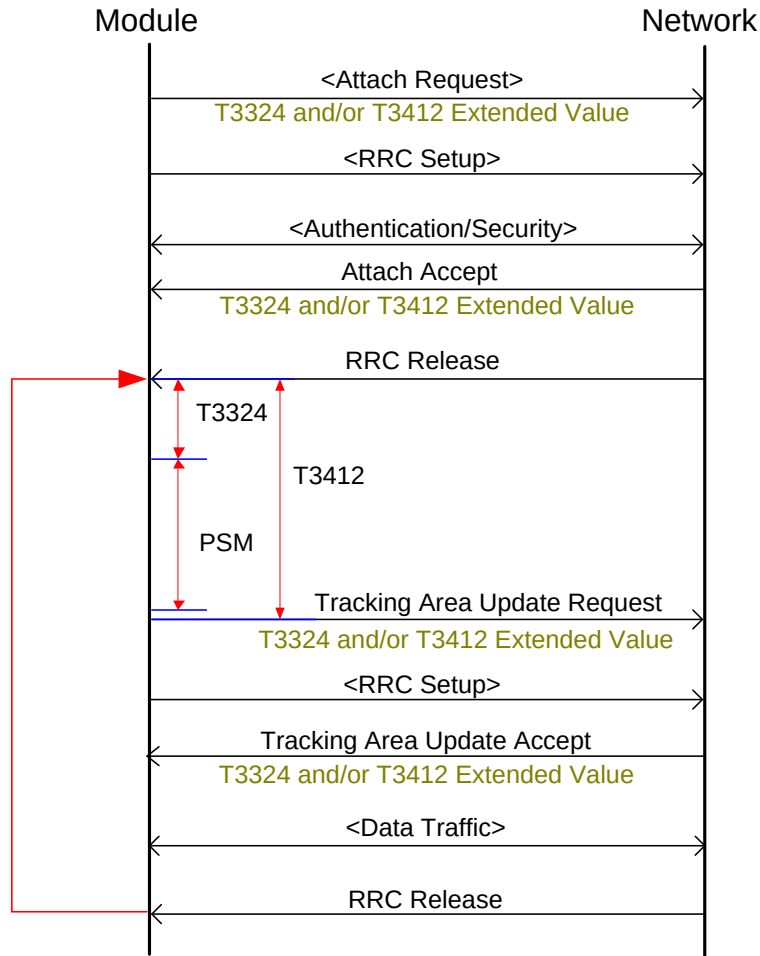


Figure 3: Waking Up the Module from PSM when T3412 Expires

4 Typical Power Consumption Cycle

The following figure shows the typical power consumption cycle of the module.

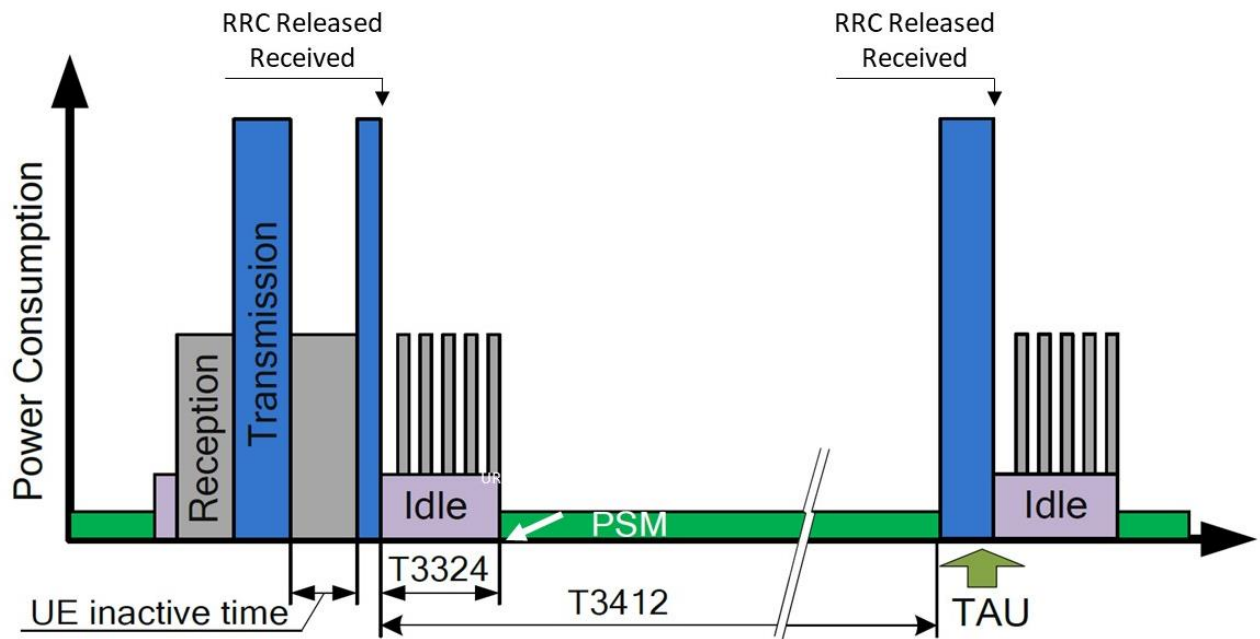


Figure 4: Typical Power Consumption Cycle

5 Appendix References

Table 1: Related Documents

Document Name
[1] 3GPP TS 23.682
[2] 3GPP TS 23.401
[3] Quectel_BG770A-GL&BG95xA-GL_AT_Commands_Manual

Table 2: Terms and Abbreviations

Abbreviation	Description
3GPP	3rd Generation Partnership Project
AP	Application Processor
DL	Downlink
eDRX	Extended Discontinuous Reception
eMTC	Enhanced Machine-Type Communication
E-UTRAN	Evolved UMTS Terrestrial Radio Access Network
MCU	Microcontroller Unit
NB-IoT	Narrow Band Internet of Things
OOS	Out Of Service
PC	Personal Computer
PDN	Packet Data Network
PSM	Power Saving Mode
RRC	Radio Resource Control

TAU	Tracking Area Update
UART	Universal Asynchronous Receiver/Transmitter
UE	User Equipment (typically, the module)
UL	Uplink
URC	Unsolicited Result Code